MA-105 Tutorial-3

Daksh Maahor

August 2025

1. **Let a function $f(x)$ be defined as:**

$$f(x) = \begin{cases} e^{-\frac{1}{x^2}} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

   Prove that f is differentiable at origin. Do higher order derivatives exist at origin? If so, find them.

2. **Let $f(x) = x^3 - 3x$. Sketch the graph of f(x). Determine the values of $a \in \mathbb{R}$ for which the equation $f(x) = a$ has 3, 1, and no real roots.**

   The points p at which $f'(p) = 0$ are called critical points, and the value of $f(x)$ at that point, $f(p)$ are called critical values. Describe what happens to the roots of $f(x) = a$ when a crosses a critical value of $f(x)$.

3. **Let $\mathbb{I}$ be a closed bounded interval, and $f : \mathbb{I} \to \mathbb{R}$ be a bounded function.**

$$M_{\mathbb{I}} = \text{lub}\{f(x) \mid x \in \mathbb{I}\}$$
$$m_{\mathbb{I}} = \text{glb}\{f(x) \mid x \in \mathbb{I}\}$$

   The value $M_{\mathbb{I}} - m_{\mathbb{I}} = \sigma(\mathbb{I})$ is called the spread of f across $\mathbb{I}$.

(a) If $\mathbb{I} \subset \mathbb{J}$ what can you say about $\sigma(\mathbb{I})$ and $\sigma(\mathbb{J})$?

(b) Fix $p \in \mathbb{I}$, consider a set of closed intervals $\mathbb{J}_n$ that all enclose p, such that length of $\mathbb{J}_n \to 0$ as $n \to \infty$. Is it necessary that for any function $f : \mathbb{I} \to \mathbb{R}$, $\sigma(\mathbb{J}_n) \to 0$ as $n \to \infty$ for the function f. If not provide a counter-example. What if f is continuous at p? Can you then say $\sigma(\mathbb{J}_n) \to 0$ as $n \to \infty$?

(c) Take an interval $[a,\ b]$ and do repeated bisections on it $[a,\ b]; [a,\ \frac{a+b}{2}]\ [\frac{a+b}{2},\ b]; [a,\ \frac{3a+b}{4}]\ [\frac{3a+b}{4},\ \frac{a+b}{2}]\ [\frac{a+b}{2},\ \frac{a+3b}{4}]\ [\frac{a+3b}{4}, b]$ etc. For simplicity, take $a = 0$ and $b = 1$.

Let $f : [0,\ 1] \to \mathbb{R}$ be a continuous function.

Let $\mathbb{I}_2 = [0,\ 1]$

Let $\mathbb{I}_3 = [0,\ \frac{1}{2}]$, $\mathbb{I}_4 = [\frac{1}{2},\ 1]$

Let $\mathbb{I}_5 = [0,\ \frac{1}{4}]$, $\mathbb{I}_6 = [\frac{1}{4},\ \frac{1}{2}]$, $\mathbb{I}_7 = [\frac{1}{2},\ \frac{3}{4}]$, $\mathbb{I}_8 = [\frac{3}{4},\ 1]$ etc.

The 2^{n-1} intervals $\mathbb{I}_{2^{n-1}+1}, \ \ldots,\ \mathbb{I}_{2^n}$ are all non overlapping and equal in length $l = \frac{1}{2^{n-1}}$.

Prove that $\lim_{n\to\infty} \sigma(\mathbb{I}_n) = 0$

If $A_n = \max\{\sigma(\mathbb{I}_j) \mid j = 2^{n-1} + 1, \ldots, 2^n\}$, prove that $\lim_{n\to\infty} \mathbb{A}_n = 0$

4. **Let $f : \mathbb{I} \to \mathbb{R}$ be a convex function on a closed interval $\mathbb{I}$. Assume $x, y, z \in \mathbb{I}$, $x < y < z$. Prove that the slope of chord joining $(x, f(x))$ and $(y, f(y))$ $\leq$ slope of chord joining $(y, f(y))$ and $(z, f(z))$.**